

ALY 3015: Final Project Report

Purpose

The purpose of the ALY 3015 Final Project Report is to consolidate your theoretical knowledge into technique and skills with practical and applicable value by analyzing a data set and presenting the findings using methods presented in this course.

Course Outcomes

This assignment is directly linked to the following key learning outcomes from the course syllabus:

CO1. Make statistical inferences about population parameters of greater business interests in order to make generalizations beyond the specific sample data. Select between ANOVA, ANCOVA, MANOVA among others.

CO2. Describe data numerically and graphically and then apply varied regression techniques to identify influential variables for business or research problems.

CO3. Integrate the application of forecast, time series and survival analysis models into business decision processes.

CO4. Classify and cluster data to identify problems and opportunities through Data Mining Techniques.

CO5. Develop an appropriate analytic model in the context of business problems.

Task

Assignment Summary

You are required to find a data set and complete an analysis of the data set using 2 – 3 methods presented in this course. You must use SPSS and R to complete the analyses.

You may work alone or with one other person. There is no difference in this assignment regardless of the number of participants in your group.

Organization

The report should include the following elements:

Introduction

Introduce the goals of the project, the data you are using, the question that needs to be answered and the methods used in your analysis. Be sure that the goals, questions and methods used fit together, meaning that questions asked reflect the goals and the methods used will answer the questions.

Your data and project topic are of your own choosing, based on your personal or professional interests. If you choose to use datasets available to the public, [UCI Machine Learning Repository](#) contains a number of preprocessed multivariate datasets. The following list (hyperlinked) contains many data sets in which you might be interested:

- [America's Best Colleges - U.S. News & World Reports](#)
- [American FactFinder](#)
- [The Baseball Archive](#)
- [The Bureau of Justice Statistics](#)
- [The Bureau of Labor Statistics](#)
- [The Bureau of Transportation Statistics](#)
- [The Census Bureau](#)
- [Data and Story Library \(DASL\)](#)
- [Data Sets, UCLA Statistics Department](#)
- [DIG Stats](#)
- [Economic Research Service, US Department of Agriculture](#)
- [Energy Information Administration](#)
- [Eurostat](#)
- [Exploring Data](#)
- [FedStats](#)
- [The Gallop Organization](#)
- [International Fuel Prices](#)
- [Journal of Statistics Education Data Archive](#)
- [Kentucky Derby Race Statistics](#)
- [National Center for Education Statistics](#)
- [National Center for Health Statistics](#)
- [National Climatic Data Center](#)
- [National Geophysical Data Center](#)
- [National Oceanic and Atmospheric Administration](#)
- [Sports Data Resources](#)
- [Statistics Canada](#)
- [StatLib---Datasets Archive](#)
- [UK Government Statistical Service](#)
- [United Nations: Cyber SchoolBus Resources](#)

Methods

You must use at least 2 methods presented in this course. Explain why your chosen methods were appropriate for the research questions and the data involved in your analysis. Be as specific as possible.

Describe any challenges you encountered during data analysis and your approach(s) to tackle the problem(s).

You must use R to complete at least one of your analyses. The other analysis/analyses can use SPSS or R.

Analysis (Body of report)

Incorporate SPSS screenshots and/or R code and the statistical output as appropriate. Use graphs, figures, charts, and tables to increase visual effects of the main points you are making based on the results.

Be sure to add your analysis and interpretation of the output and the significance of the results.

Conclusions & Limitations

Use the conclusion to summarize the analysis and findings. Present an answer to the questions posed in the introduction.

Identify any research design or methodology limitations that may affect or influence the interpretation of the findings from your research.

Format

Your Final Project report should follow correct APA format and can be up to 10 pages double-spaced (excluding title page and references page).

Final Report Rubric

Category	Meets Standards	Approaching Standards	Below Standards
Introduction	Clearly and briefly introduces the goals of the project, the data used, the questions that need to be answered and the methods used in the analysis. The goals, questions and methods outlined are consistent with one another.	Introduces the goals of the project, the data used, the question that needs to be answered and the methods used in the analysis but is too wordy or the goals, questions and methods are inappropriate.	Does not introduce project goals, the data used, project questions, and/or methods, or they are not aligned with one another.
Methods	Uses at least 2 methods as identified in the introduction in the analysis. Clearly and fully explains the methods and data used in the analysis and provides valid reasoning.	Uses 2 methods as identified in the introduction in the analysis. Explains the methods and data used in the analysis, but is too cursory or vague, or reasoning for methods is not sufficiently coherent.	Does not identify 2 methods used in the analysis. Does not explain the reasoning for the data or methods used in the analysis.

Category	Meets Standards	Approaching Standards	Below Standards
Analysis	Incorporates SPSS screenshots and output and/or R code and the outputs as appropriate. Provides accurate, detailed analysis of the output focusing on significance of results. Uses visualizations to make major points	Incorporates SPSS screenshots and output and/or R code and outputs. Provides generalized, mostly accurate analysis of the outputs. Uses visualizations to address some major points.	Includes none to few SPSS screenshots and output and/or R code and outputs. Provides unclear or inaccurate statements in the analysis. Uses few or inappropriate visualizations or they do not pertain to main points.
Conclusions & Limitations	Wraps up the findings in a conclusion that provides an answer to the question(s) posed in the introduction. Accurately identifies any research design or methodology limitations that may affect or influence the interpretation of the findings from the research.	Wraps up the findings in a conclusion that provides general answers to the question posed in the introduction. Describes vague or general research design or methodology limitations that may affect or influence the interpretation of the findings from the research.	Does not wrap up findings in a conclusion or does not provide an answer to the question. Does not identify any research design or methodology limitations or identifies limitations that do not pertain to or affect/influence the interpretation of the findings from the research.
Data Visualizations	Data visualizations are appropriate for the level and type of analysis. Uses graphs, figures, charts, and tables to increase visual effects of the main points	Data visualization are useful for the level and type of analysis, but graphs, figures and tables do not clearly communicate significance	Data visualizations are used minimally or not at all. If graphs, figures and tables are used, it is unclear what they are intended to communicate or why or they are inappropriate for the purpose.

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Category	Meets Standards	Approaching Standards	Below Standards
	being made based on the results.	of the results to the reader—or other visualizations would have been more appropriate.	
Report: Writing Mechanics, Title Page, & References	There are no noticeable errors in grammar, spelling, and punctuation; and completely correct usage of title page, citations, and references. The report is no more than 10 pages, double-spaced (excluding title page and references page).	There are very few errors in grammar, spelling, and punctuation; and completely correct usage of title page, citations, and references. The report is no more than 10 pages, double-spaced (excluding title page and references page).	There are more than five errors in grammar, spelling, and punctuation; or the usage of title page, citations, and references are incomplete; or The report is more than 10 pages, double-spaced (excluding title page and references page).